

Optimal Evasion of Surface-to-Air Missile Detection Networks

A Computational Geometry, Graph-Theoretic, and Strategic Analysis

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Abstract

The problem of navigating an aircraft through a region protected by multiple Surface-to-Air Missile (SAM) systems is examined from the perspectives of computational geometry, graph theory, operations research, military science, economics of warfare, political science, international relations, artificial intelligence, and algorithm design. The objective is to determine a shortest feasible path from an origin point to a destination while avoiding all SAM detection zones. The resulting formulation leads to a shortest-path problem in a continuous free-space manifold. A visibility-graph algorithm based on tangencies between circular threat regions is presented together with mathematical proofs of correctness, complexity analysis, strategic implications, and implementation considerations.

The paper ends with “The End”

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1 Introduction

Modern integrated air-defense systems create denial regions in which adversarial aircraft face destruction upon detection.

Suppose there are N SAM sites

$$S_i = (x_i, y_i),$$

each possessing a circular detection radius

$$R_i > 0.$$

The aircraft begins at

$$O = (O_x, O_y)$$

and seeks to reach

$$D = (D_x, D_y)$$

while remaining outside all detection zones.

The problem combines:

- Computational geometry;
- Graph theory;
- Operations research;
- Military planning;
- Artificial intelligence path planning;
- Geospatial optimization.

2 Mathematical Formulation

The forbidden region associated with site i is

$$\mathcal{D}_i = \{(x, y) : (x - x_i)^2 + (y - y_i)^2 \leq R_i^2\}.$$

The total denied airspace is

$$\mathcal{D} = \bigcup_{i=1}^N \mathcal{D}_i.$$

The free space is

$$\mathcal{F} = \mathbb{R}^2 \setminus \mathcal{D}.$$

The objective is to find a continuous curve

$$\gamma : [0, 1] \rightarrow \mathcal{F}$$

such that

$$\gamma(0) = O, \quad \gamma(1) = D,$$

while minimizing

$$L(\gamma) = \int_0^1 \|\gamma'(t)\| dt.$$

3 Geometric Properties of Optimal Paths

Theorem 1. *Any shortest collision-free path among circular obstacles consists solely of straight-line segments and circular boundary arcs.*

Proof. Suppose a shortest path contains a smooth interior segment that is neither straight nor on an obstacle boundary.

A non-straight interior curve can be replaced by its chord.

The chord is shorter by the triangle inequality and remains within free space for sufficiently small perturbations.

This contradicts optimality.

Therefore all interior portions must be straight.

Whenever the path touches a circular obstacle, it must remain tangent to the obstacle boundary. Otherwise a local shortening exists.

Hence the path consists exclusively of tangent segments and circular arcs. □

Corollary 1. *The optimal path is completely characterized by obstacle tangencies.*

4 Visibility Graph Construction

4.1 Vertices

Vertices are generated from:

1. Origin point O ;
2. Destination point D ;
3. Tangent points from O to every circle;
4. Tangent points from D to every circle;
5. Internal circle-circle tangencies;
6. External circle-circle tangencies.

4.2 Edges

Two types of edges exist.

4.2.1 Visibility Edges

Two vertices u, v are connected if the segment joining them avoids every obstacle.

Weight:

$$w(u, v) = \|u - v\|.$$

4.2.2 Boundary Arc Edges

For points lying on the same obstacle,

$$w(u, v) = R_i \Delta\theta,$$

where $\Delta\theta$ denotes angular separation.

5 Circle Geometry

5.1 Point-Circle Tangency

Let

$$P = (p_x, p_y)$$

be outside circle

$$C = (c_x, c_y, R).$$

Distance:

$$d = \sqrt{(p_x - c_x)^2 + (p_y - c_y)^2}.$$

Tangency exists iff

$$d > R.$$

The tangent angle satisfies

$$\beta = \arccos\left(\frac{R}{d}\right).$$

Hence tangent points are

$$T_{\pm} = (c_x + R \cos(\alpha \pm \beta), c_y + R \sin(\alpha \pm \beta)),$$

where

$$\alpha = \tan^{-1}\left(\frac{p_y - c_y}{p_x - c_x}\right).$$

5.2 Circle-Circle Tangencies

Let circles be

$$C_1 = (x_1, y_1, r_1),$$

$$C_2 = (x_2, y_2, r_2).$$

Define

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

Tangencies are generated by solving

$$\|P_1 - P_2\| = d,$$

under equal-normal constraints.

The resulting formulas produce both external and internal tangent pairs.

6 Graph-Theoretic Formulation

Construct graph

$$G = (V, E).$$

- Vertices correspond to tangency points.
- Edges correspond to visible segments or arcs.

The shortest path becomes

$$\min_{\pi} \sum_{e \in \pi} w(e).$$

7 Shortest Path Algorithm

Dijkstra's algorithm is applied.

7.1 Pseudo-Code

Generate all tangency vertices

For every pair of vertices:

 If visible:

 Add straight edge

For every obstacle:

 Sort tangent points by angle

 Add circular arc edges

Run Dijkstra(0,D)

Return shortest path

8 Complexity Analysis

Assume:

N = number of SAM sites.

Circle-circle tangencies:

$$O(N^2).$$

Visibility graph vertices:

$$V = O(N^2).$$

Visibility testing:

$$O(V^2N).$$

Dijkstra:

$$O(E \log V).$$

Total complexity:

$$O(N^5)$$

in the worst theoretical case.

Practical performance is substantially better because visibility constraints eliminate many edges.

9 Artificial Intelligence Perspective

The problem belongs to a broader class of motion-planning tasks.

Related AI algorithms include:

- A* Search;
- Probabilistic Roadmaps;
- Rapidly Exploring Random Trees;
- Reinforcement Learning;

- Monte Carlo Tree Search.

For dynamic SAM environments, a Markov Decision Process may be formulated.
State:

$$s_t = (x_t, y_t).$$

Action:

$$a_t = (\Delta x, \Delta y).$$

Reward:

$$R_t = -\text{distance} - \lambda \cdot \text{risk}.$$

10 Statistical Considerations

Suppose detection is probabilistic.

Detection probability for site i :

$$P_i(r) = 1 - e^{-\lambda_i r}.$$

Survival probability along route γ :

$$P_{\text{survival}} = \prod_i (1 - P_i).$$

Expected mission value:

$$EV = P_{\text{survival}}V - C.$$

Here:

- V = mission value,
- C = operational cost.

11 Economics of Warfare

Let

$$M = \text{value of target}.$$

Aircraft cost:

$$A.$$

Mission success probability:

$$p.$$

Expected military utility:

$$U = pM - (1 - p)A.$$

A rational commander launches the mission iff

$$U > 0.$$

SAM deployment itself has economic cost

$$C_{SAM}.$$

An optimal defender chooses

N

and

R_i

to maximize

$$\text{Deterrence} - C_{SAM}.$$

12 Political Science and International Relations

Air-defense networks alter strategic equilibria.

Examples include:

- Anti-access/Area-denial (A2/AD);
- Deterrence by denial;
- Escalation management;
- Strategic signaling.

A state's effective sovereignty over airspace increases with coverage fraction

$$\phi = \frac{\text{Protected Area}}{\text{Total Area}}.$$

Higher ϕ generally raises the cost of military intervention.

13 Military Operational Interpretation

The path-planning problem models:

- Strike missions;
- Reconnaissance flights;
- Cruise missile routing;
- Unmanned aerial vehicle navigation;
- Penetration of Integrated Air Defense Systems.

The shortest safe route corresponds to a minimum-risk ingress corridor.

14 Proof of Correctness

Theorem 2. *The visibility-graph algorithm returns a globally shortest feasible path.*

Proof. Every shortest path consists only of tangencies and obstacle arcs.

All such tangencies are represented as vertices.

Every feasible straight segment between tangencies appears as a graph edge.

Every feasible obstacle traversal appears as an arc edge.

Therefore every optimal geometric path corresponds to a graph path.

Conversely every graph path corresponds to a feasible geometric path.

Thus shortest-path optimization over the graph is equivalent to shortest-path optimization in continuous space.

Dijkstra's algorithm is optimal on graphs with nonnegative weights.

Hence the returned route is globally optimal.

□

15 Implementation Summary

Table 1 summarizes the principal computational components.

Table 1: Algorithm Components

Component	Purpose
Tangency Generator	Candidate vertices
Visibility Test	Feasible straight edges
Arc Generator	Boundary traversal edges
Graph Builder	Construct visibility graph
Dijkstra Search	Optimal route computation
Path Reconstruction	Recover flight plan

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Glossary

SAM Surface-to-Air Missile system.

Visibility Graph Graph whose vertices are tangencies and whose edges are unobstructed connections.

Free Space Region outside all SAM detection zones.

Tangency Point Point where an optimal route touches an obstacle boundary.

A2/AD Anti-Access / Area-Denial strategy.

Dijkstra Algorithm Optimal shortest-path algorithm for nonnegative edge weights.

MDP Markov Decision Process.

Expected Utility Probability-weighted strategic value.

The End